IAS PRELIMS/MAINS

Ecology & Environment

A COMPREHENSIVE COVERAGE

MODULE - 43

CLIMATE CHANGE - 4

7 CLIMATE CHANGE IMPACT ON INDIAN AGRICULTURE

- Agriculture and allied sectors exhibit high sensitivity to climatic changes.
- Temperature, precipitation and humidity play a fundamental role affecting the productivity in agriculture, livestock and fisheries sectors.
- Any change in these factors not only leads to adverse impacts on food security and nutrition, but also affects the livelihood of millions dependent on the farming sector.
- The overall impact of climate change on our food production and economy is expected to be high since the agriculture and allied sector still accounts for a large share of Gross Domestic Product (GDP) and employment.

Learning Space

7 CLIMATE CHANGE IMPACT ON INDIAN AGRICULTURE

IMPACTS

- Since most of the cultivated land in India is rainfed and agricultural production is heavily dependent on the monsoons, agriculture productivity and the well-being of the Indian farmer are very sensitive to climatic variability.
- Rainfall during the monsoon season (June-September) is the key element of the Indian climate and continues to be the primary source of water for the large rainfed agricultural regions in the country.
- The changes in temperatures influences plant physiological conditions viz., respiration, water requirement and growth, thereby affecting yields.
- Extreme weather conditions due to climate change such as floods, droughts, heat and cold waves, flash floods, cyclones, hail storms, etc. are constant hazards for agricultural production.

IMPACTS

- Even small variations in weather conditions during critical phases of crop development have substantial impact on yields.
- Most crop studies have predicted a decrease in the yield of crops with an increase in temperature. Further, adverse temperature and moisture conditions affect the quality of food grains.
- The damage to crops caused by pests, pathogens and weeds increases due to climate change. Change in climate is likely to bring about a change in the population dynamics, growth and distribution of insects and pests thereby, upsetting crop-pest balance.
- Drought conditions would benefit pathogens and insects due to decrease in plant defense system. These changes could lead to enormous crop losses.
- Changes in precipitation patterns and amounts, and variation in temperature may degrade soil quality, reduce soil moisture content and affect microbial diversity, which in turn affect crop growth.

IMPACTS

- An increase in temperature also leads to increased evapo-transpiration, thereby reducing the soil moisture.
- At some places, increased surface temperature coupled with reduced rainfall may lead to accumulation of salts in upper soil layers.
- Similarly, a rise in sea level associated with increased temperature may lead to salt-water ingression in the coastal lands, making them unsuitable for agriculture.
- Impacts of climate change on livestock will be felt in the form of decrease in feed intake, etc. Indirect impacts would be observed in the form of reduction in grazing land and water availability, decline in available cattle feed, emergence of new diseases, etc.
- Heat stress would impact profitability in dairying due to lower feed intake, milk production and reproduction.
- The livestock will also be exposed to the risks associated with extreme events.
- Changes in the climate would also affect the fisheries sector in many ways.

FIGHTING CLIMATE CHANGE – MITIGATION AND ADAPTATION

- There are two main tools to fight climate change - Mitigation and Adaptation.
- Mitigation addresses the root cause, by reducing greenhouse gas emissions, while adaptation seeks to lower the risks posed by climatic changes.
- Both approaches will be necessary, because even if emissions are dramatically decreased in the next decade, adaptation will still be needed to deal with the global changes that have already been set in motion.
- Adaptation measures include large-scale infrastructure changes – such as building defences to protect against sea-level rise or going for intensive mangrove plantation– as well behavioural shifts such as individuals using less water, farmers planting different crops etc.

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FIGHTING CLIMATE CHANGE – MITIGATION AND ADAPTATION

- The IPCC describes vulnerability to climate change as being determined by three factors
  1. Exposure to hazards (such as reduced rainfall),
  2. Sensitivity to those hazards (such as an economy dominated by rain-fed agriculture),
  3. The capacity to adapt to those hazards (for example, whether farmers have the money or skills to grow more drought-resistant crops).
     - ✓ Adaptation measures can help reduce vulnerability.

ipcc

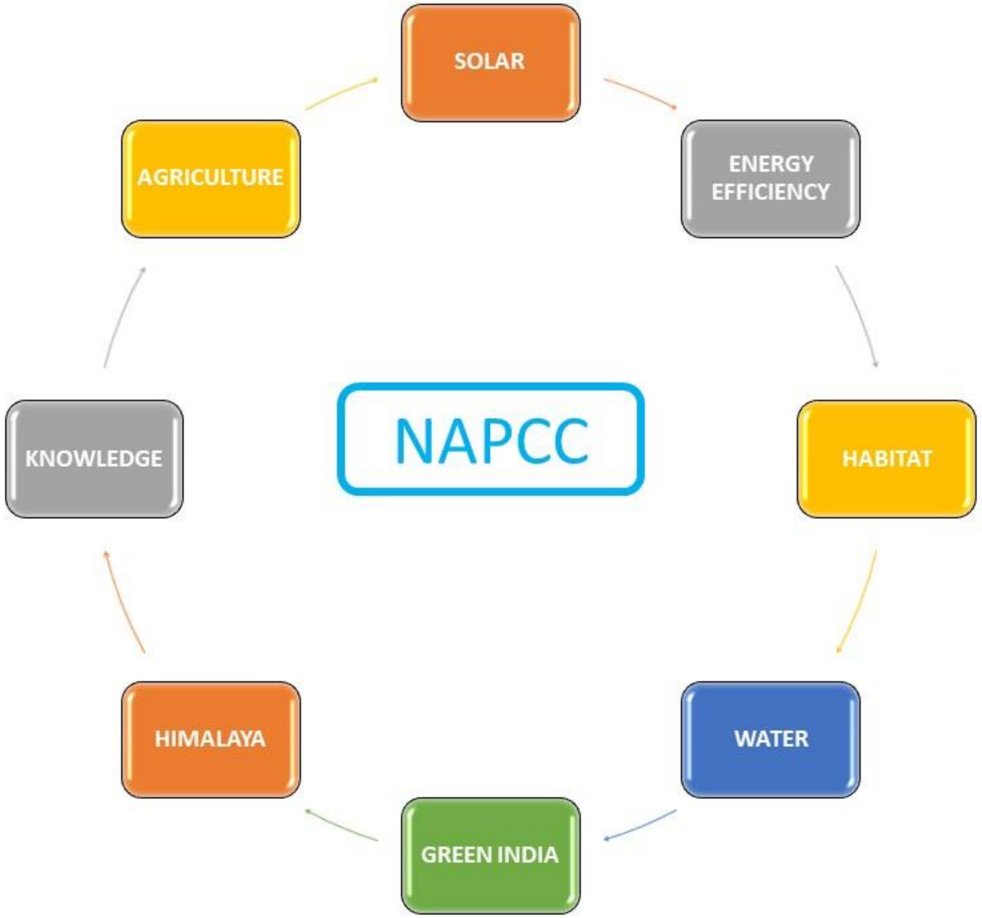
WMO UNEP

INTERGOVERNMENTAL PANEL ON climate change

7 CLIMATE CHANGE IMPACT ON INDIAN AGRICULTURE

INDIA'S NATIONAL ACTION PLAN ON CLIMATE CHANGE (NAPCC)

- NAPCC was launched in June 2008 by PM's Council on Climate Change.
- The NAPCC identifies measures that promote development, while also effectively addressing climate change.
- The NAPCC consists of eight national missions.
- The focus of NAPCC will be on promoting understanding of climate change, adaptation and mitigation, energy efficiency and natural resources conservation.



WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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